A sum of squares for the Hartogs-triangle polynomials

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Abstract

We prove Edholm–Mathew's Conjecture 4.20: for all coprime positive integers $m > n$, the polynomial $Q_{m,n}$ has no zeros on the unit circle and has exactly $m - n$ zeros in the punctured unit disc, counted with multiplicity. We express its centered Laurent polynomial as a sum of squared absolute values of explicit integer polynomials. Two of these polynomials differ by a monomial times $1 - z$, which rules out simultaneous vanishing. The established Bergman-kernel numerator formula is the only input from the source.

Task 1. Full solution

Let $m > n \geq 1$, $\gcd(m, n) = 1$, and $k = m - n$. Define

$$D_m(j) = \max\{0, m - |j - (m-1)|\} \qquad (j \in \mathbb{Z}).$$

Almughrabi's kernel formula, recalled in [1, Proposition 1.10, equation (1.11)], gives the numerator

$$P_{m,n}(s,t) = \sum_{\beta_1,\beta_2 \in \mathbb{Z}} D_m(\beta_1) D_m(m\beta_2 + n\beta_1 + m + n - 1 - 2mn) s^{\beta_1} t^{\beta_2}. \tag{1}$$

Only finitely many terms are nonzero. The polynomial in the question is

$$Q_{m,n}(z) = z^{1-2n} P_{m,n}(z, z).$$

We use the numerator (1) with precisely the source's normalization. For $0 \leq c < m$, introduce

$$H_c(z) = \sum_{a=0}^{m-1} z^{\,a - \lfloor (na+c)/m \rfloor}. \tag{2}$$

These are polynomials with nonnegative integer coefficients: $na + c < m(a+1)$ implies $\lfloor (na + c)/m \rfloor \leq a$. Each $H_c(1) = m$.

Theorem 1. *For $z \neq 0$, the following Laurent-polynomial identity holds:*

$$\boxed{z^{-k} Q_{m,n}(z) = \sum_{c=0}^{m-1} H_c(z) H_c(z^{-1}).} \tag{3}$$

In particular, for every $|z| = 1$,

$$z^{-k} Q_{m,n}(z) = \sum_{c=0}^{m-1} |H_c(z)|^2 > 0. \tag{4}$$

Consequently $Q_{m,n}$ has exactly k zeros in $\mathbb{D} \setminus \{0\}$ and k outside the closed unit disc, counted with multiplicity. This proves Conjecture 4.20 in full.

Centering the known numerator

Write $W_m(u) = m - |u|$ for $|u| < m$ and $W_m(u) = 0$ otherwise. In (1), set

$$u = \beta_1 - (m-1), \qquad v = m\beta_2 + n\beta_1 + n - 2mn.$$

The second argument of D_m is $v + (m-1)$ and

$$v - nu = m(\beta_2 - n).$$

Thus this change of variables is a bijection between integer pairs (β_1, β_2) and pairs (u, v) satisfying $v \equiv nu \pmod m$. On the diagonal, the exponent after centering is

$$\beta_1 + \beta_2 - (m+n-1) = \frac{ku+v}{m}.$$

Therefore, with $R(z) = z^{-k}Q_{m,n}(z)$,

$$R(z) = \sum_{\substack{|u|,|v|<m \\ v\equiv nu \pmod m}} W_m(u)W_m(v)z^{(ku+v)/m}. \tag{5}$$

The exponents are integers. Their absolute values are bounded by $(k+1)(m-1)/m < k+1$, hence by k. The pair $(u, v) = (m-1, k)$ is admissible and contributes positively to z^k. Simultaneous negation of u, v preserves the weights in (5), so $R(z) = R(z^{-1})$ and the coefficient of z^{-k} is also positive. This also proves directly that $Q_{m,n}$ has degree $2k$, nonzero constant term, and

$$Q_{m,n}(z) = z^{2k}Q_{m,n}(z^{-1}), \tag{6}$$

the palindromic property previously established in [1].

An integer sum-of-squares certificate

For $a \in \{0, \dots, m-1\}$ let $b_c(a)$ be the least nonnegative residue of $na + c$ modulo m. Then

$$e_c(a) := a - \left\lfloor \frac{na+c}{m} \right\rfloor = \frac{ka + b_c(a) - c}{m}. \tag{7}$$

Expand the right hand side of (3), using (2) and (7):

$$\sum_{c=0}^{m-1} \sum_{a,a'=0}^{m-1} z^{\{k(a-a')+b_c(a)-b_c(a')\}/m}. \tag{8}$$

Every ordered pair of grid points $(a, b), (a', b')$ in $\{0, \dots, m-1\}^2$ with

$$b - na \equiv b' - na' \pmod m \tag{9}$$

appears exactly once in (8), because (9) specifies a unique common residue c, and then $b = b_c(a)$, $b' = b_c(a')$.

Fix $u = a - a'$ and $v = b - b'$. There are $W_m(u)$ ordered pairs (a, a') with difference u, and independently $W_m(v)$ ordered pairs (b, b') with difference v. All of them satisfy (9) exactly when $v \equiv nu \pmod m$. Their common exponent in (8) is $(ku+v)/m$. Grouping (8) by these two differences gives (5), which proves (3). Because all coefficients of H_c are real, $H_c(z^{-1}) = \overline{H_c(z)}$ on $|z| = 1$.

Why the sum is strictly positive

Coprimality gives a unique $a_* \in \{1, \dots, m-1\}$ such that $na_* \equiv -1 \pmod{m}$. Put

$$q_* = \frac{na_* + 1}{m}, \qquad r = a_* - q_* = \frac{ka_* - 1}{m} \geq 0.$$

As c changes from 0 to 1 in (2), the floor increases by one only at $a = a_*$. The old exponent there is $r+1$ and the new exponent is r. Hence the exact identity

$$\boxed{H_1(z) - H_0(z) = z^r(1-z).} \tag{10}$$

If $|z| = 1$ and $z \neq 1$, (10) shows that $H_0(z)$ and $H_1(z)$ cannot both vanish. If $z = 1$, every $H_c(1) = m$. Therefore (3) is strictly positive at every point on the unit circle. In fact,

$$R(z) \geq \frac{1}{2}|1-z|^2 \quad (|z| = 1), \qquad R(1) = m^3, \tag{11}$$

since $|H_0|^2 + |H_1|^2 \geq \frac{1}{2}|H_1 - H_0|^2$.

This proves nonvanishing. Finally (6) pairs each nonzero root z with $1/z$, with equal multiplicity. There are $2k$ roots in total, none at zero or on the circle. Exactly k are inside the punctured unit disc and k are outside, completing the proof of Theorem 1.

For example, $(m, n) = (3, 2)$ gives $H_0 = 1 + 2z$, $H_1 = 2 + z$, $H_2 = 3$, and

$$z^{-1}Q_{3,2}(z) = 19 + 4(z + z^{-1}) = |1 + 2z|^2 + |2 + z|^2 + 9 \qquad (|z| = 1).$$

The argument above applies without any restriction on $m - n$.

Consequence for the Bergman kernel

The known formula for the Bergman kernel on $\mathbb{H}_{m/n} = \{(z_1, z_2) : |z_1|^{m/n} < |z_2| < 1\}$ is

$$K_{m/n}(z, w) = \frac{P_{m,n}(s, t)}{m\pi^2(1-t)^2(t^n - s^m)^2}, \qquad s = z_1\overline{w_1}, \quad t = z_2\overline{w_2}. \tag{12}$$

Take a root $\xi = \rho e^{i\theta}$ of $Q_{m,n}$ in the punctured disc, and set $z = (\sqrt{\rho}e^{i\theta}, \sqrt{\rho}e^{i\theta})$, $w = (\sqrt{\rho}, \sqrt{\rho})$. Since $m/n > 1$ and $0 < \rho < 1$, both points lie in $\mathbb{H}_{m/n}$ and give $s = t = \xi$. The numerator in (12) is zero, while its denominator is nonzero: $\xi \neq 1$ and $\xi^n - \xi^m = \xi^n(1 - \xi^k) \neq 0$. Thus the kernel has zeros for every rational $m/n > 1$.

The numerator formula and the earlier palindromic and special-case results are due to the cited authors. The new contribution is the explicit integer certificate (3) together with the obstruction (10) to simultaneous vanishing.

Tasks 2–5. Proposed follow-ups

Task 2. Determine the minimum of $z^{-k}Q_{m,n}(z)$ on the unit circle and obtain sharp bounds, in terms of m, n, for how close the roots of $Q_{m,n}$ can lie to that circle.

Task 3. Determine the number of negative real roots of $Q_{m,n}$ and whether all its complex roots are simple, including the dependence on the residue class of n for fixed $m - n$.

Task 4. Investigate whether analogous residue-class sum-of-squares certificates control boundary zeros of Bergman numerators on other two-dimensional monomial quotient domains.

Task 5. Decide whether the Bergman kernel of $\mathbb{H}_\gamma$ has zeros for every irrational $\gamma \in (1, 2)$, with quantitative control of zeros under rational approximation.

Verification. Four exact groups pass: the two source formulas and the sum-of-squares identity for all 773 coprime pairs with $m \leq 50$; independent enumeration of 136,420 admissible ordered quadruples for $m \leq 16$; exact Sturm counts for 199 cosine polynomials with $m \leq 25$; and 12 larger cases up to $(m, n) = (10007, 10002)$. No floating-point arithmetic is used. These checks supplement the proof. Review is by the authoring agent.

References

[1] L. D. Edholm and V. T. Mathew, *Arithmetic properties and zeros of the Bergman kernel on a class of quotient domains*, Math. Z. **313**, article 43 (2026), doi:10.1007/s00209-026-04059-8; arXiv:2505.20489v3. The target is Conjecture 4.20. Proposition 1.10 attributes (1) to R. Almughrabi, *Bergman kernels of two dimensional monomial polyhedra*, Complex Anal. Oper. Theory **17**, article 103 (2023).